

NSGA-II – Mathematical Formulation

Gati Shakti Vishwavidyalaya

1 What NSGA-II Does

NSGA-II optimizes three objectives simultaneously, producing a Pareto front of 60 non-dominated solutions representing different operational trade-offs. *Code: src/algorithms/nsga2.py, function solve_nsga2()*

2 Three Objectives (all minimize)

$$f_1 = D_{head} \quad (\text{deadhead km})$$

$$f_2 = N_{units} \quad (\text{units used})$$

$$f_3 = P_{uncov} + 100 \cdot V_{time} \quad (\text{demand} + \text{timing penalty})$$

Computed in: evaluate_chromosome() in nsga2.py

3 Domination Rule

Solution A dominates B if:

$$f_i(A) \leq f_i(B) \quad \forall i \in \{1, 2, 3\} \quad \text{and} \quad f_j(A) < f_j(B) \quad \text{for some } j$$

Code: dominates() function in nsga2.py

4 Pareto Rank Assignment

Rank 0 = solutions no one dominates (Pareto front).

Rank 1 = dominated only by Rank 0 solutions. And so on. *Code: fast_non_dominated_sort() in nsga2.py*

5 Crowding Distance

For solutions at the same rank, prefer those in less crowded regions:

$$cd_i = \sum_{k=1}^3 \frac{f_k(i+1) - f_k(i-1)}{f_k^{max} - f_k^{min}}$$

Higher crowding distance = more unique trade-off = preferred in selection. *Code: compute_crowding_distance() in nsga2.py*

6 Selection

Binary tournament: compare two solutions on (rank, crowding distance). Lower rank wins. Ties broken by higher crowding distance. *Code: binary_tournament() in nsga2.py*

7 Crossover and Mutation

SBX crossover ($\eta = 15$, $p = 0.9$): smooth blend of parent gene values. Polynomial mutation ($\eta = 20$, $p = 1/T$): perturb individual genes. After both: call `repair_chromosome()` from `src/Utils/feasibility.py`.
Code: `sbx_crossover()`, `polynomial_mutation()` in `nsga2.py`

8 Generation Loop

Parents (60) + children (60) = 120 combined.
Sort all 120 by rank then crowding distance.
Keep best 60 for next generation. *Code: `generation loop` in `solve_nsga2()`*

9 Why NSGA-II Has Scaling Issues

Splitting 60 individuals across 3 objectives means no single objective gets full optimization pressure. At 120 trips: 39 timing violations. The 3-objective pressure sacrifices constraint satisfaction at scale.